

Sun Minghui

15144224510

2023000416@mails.cust.edu.cn

Changchun University of Science and Technology, China

Summary

Third-year undergraduate student majoring in Applied Physics at Changchun University of Science and Technology, interested in building reliable, physics-informed models for mechanical structure design, solid-state lasers, micro-LEDs, and PINNs.

Education

B.S., Changchun University of Science and Technology, Major in Applied Physics

- Ranked 1/47

Research Experience

Modeling and Optimization of Tunable Solid-State Lasers Based on Seed Injection

Locking, Project Leader

CUST

Jan. 2025 to Present

- Developed physics-based models for seed injection locking in tunable 2- μm solid-state lasers, enabling systematic analysis of seed-cavity mode coupling dynamics.
- Investigated nonlinear multi-wavelength tuning behavior and thermo-optical coupling via analytical modeling and numerical simulations, improving frequency stability and spectral purity.
- Optimized key system parameters and predicted minimum injection power to support experimental design and performance optimization.

Physics-Informed Neural Network Modeling for Sodium-Ion Battery Systems

Project Leader

CUST

Feb. 2024 to Feb. 2025

12 months

- Built a physics-informed neural network (PINN) framework for the sodium-ion battery single-particle model (SPM) by embedding electrochemical governing equations and boundary conditions.
- Proposed an attention-enhanced residual-coupled PINN (RCA-PINN) architecture to reduce voltage prediction errors and improve computational efficiency under high-rate conditions.
- Achieved >50% error reduction with ~ 0.01 s/step inference time, enabling real-time state estimation for battery management systems.

Operator-Consistent Physics-Informed Learning for Wafer Thermal Reconstruction in

Lithography, Research Assistant

CUST

Jan. 2025 to Oct. 2025

9 months

- Developed an operator-consistent physics-informed learning framework to reconstruct wafer temperature fields under sparse and noisy measurements in lithography processes.
- Implemented physics-based constraints and operator learning modules to improve generalization and stability across varying boundary conditions and thermal diffusion regimes.
- Conducted numerical experiments and ablation studies to validate reconstruction accuracy and physical consistency.

Additional Experience And Awards

Granted Invention Patent ×1

Invention Patents under Review ×2 (including 1 as **First Inventor**)

Utility Model Patents under Review ×3 (including 1 as **First Inventor**)

Provincial Government Scholarship (Highest Provincial-Level Award), 2024–2025

President's Scholarship (University-wide Top Award), CUST — Spring 2024

Merit Student (University-wide Honor), CUST — 2023–2024

First-Class Scholarship ×2, CUST— Fall 2023 / Spring 2025

Second-Class Scholarship, Changchun University of Science and Technology— Fall 2024

National Undergraduate Innovation Training Program (Key Project), Project Leader, 2025

Provincial-Level Undergraduate Innovation Project, Core Member, 2024 (Project Initiation)

Provincial-Level Undergraduate Innovation Project, Core Member, 2025 (Project Completion)

National :

China Undergraduate Engineering Practice and Innovation Competition: Second Prize, 2025

China Robot and Artificial Intelligence Competition: Third Prize, 2025

Global Campus Artificial Intelligence Algorithm Competition: Excellence Award, 2025

Provincial / Regional Level :

Jilin Provincial Undergraduate Engineering Practice and Innovation Competition: Grand Prize, 2025

National Undergraduate Optoelectronic Design Competition : Second Prize

“Saibo Cup” Intelligent Robot Creative Competition : Second Prize, 2025

China Robot and Artificial Intelligence Competition : Second Prize

Global Campus Artificial Intelligence Algorithm Competition: Second Prize, 2025

Publications (Under Review)

• **Minghui Sun**, et al. *Residual-channel attention physics-informed neural network for surrogate modeling of a sodium-ion battery single-particle model.s.* **Journal of Power Sources**, 682 (2026) 240366. **Published (First Author).**

• Chunyang Wang, **Minghui Sun**, et al. *2 μm dual-wavelength Tm:YAG laser based on a $\text{MoS}_2/\text{Ti}_3\text{C}_2\text{T}_x$ saturable absorber.* **Infrared Physics & Technology**, 156 (2026) 106586. **Published (Second Author).**

• Ze Tao, Yuxi Jin, Ke Xu, **Minghui Sun**, et al. *Operator-Consistent Physics-Informed Learning for Wafer Thermal Reconstruction in Lithography.* *Journal of Computational Physics* — Under Review (Fourth Author). Manuscript ID: JCOMP-D-25-02717.

Technologies

Programming: Python (PyTorch), MATLAB

Modeling & Simulation: PINNs, numerical simulation, optimization

Tools: SolidWorks, KeyShot, LabVIEW, Overleaf (LaTeX)